Project IQ - External Libraries & Tools Inventory

Document Version: 1.0

Application: Project IQ - Rasa-based IT Support Chatbot

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Backend Dependencies

Core Framework Stack

Rasa (v3.6.20)

- Purpose: Open-source conversational AI framework for building the chatbot
- Usage:
- NLU (Intent/Entity classification)
- Core dialogue management
- Policy-based response generation
- Key Files: domain.yml, nlu.yml, stories.yml, rules.yml, data/
- Port: 5005 (Rasa Core server)
- Dependencies: TensorFlow/PyTorch, Sanic (async web framework)

Rasa-SDK (v3.6.2)

- Purpose: SDK for developing custom actions in Python
- Usage:
- Execute custom business logic (GLPI integration, KB search, etc.)

- ActionGlpiLogin, ActionCreateGlpiTicket, ActionSearchKnowledgeBase, etc.
- Key Files: actions/actions.py, actions/*.py (all action handlers)
- Port: 5055 (Action Server)

Pydantic (v1.10.9)

- Purpose: Data validation and settings management
- Usage: Validate request/response schemas, configuration objects
- Key Files: Config models in actions/config_manager.py

Typing-extensions (v4.5.0)

- Purpose: Backport of Python typing features for compatibility
- Usage: Type hints in codebase

Packaging (v20.9)

- Purpose: Core utilities for handling package versions
- Usage: Version comparison, dependency management

Web Framework & API

Flask (v3.0.0)

- Purpose: Lightweight Python web framework for Admin API
- Usage:
- REST API endpoints for admin dashboard
- Static file serving (HTML, CSS, JS)
- Configuration management routes
- Analytics endpoints
- Key Files: actions/admin_api.py
- Port: 80/8181 (Admin API)
- Endpoints: /admin/*, /themes/*, /config/*, /health/*

Flask-CORS (v4.0.0)

- Purpose: Cross-Origin Resource Sharing support
- Usage: Allow frontend to call backend from different origin
- Key Files: actions/admin_api.py (CORS initialization)

Werkzeug (v3.0.1)

- Purpose: WSGI utilities and HTTP handling
- Usage: HTTP request/response handling (used by Flask)
- Dependencies: Flask dependency

Database & ORM

PostgreSQL (via psycopg2-binary v2.9.9)

- Purpose: Production-grade relational database
- Usage:
- Conversation tracker store (messages, sessions)
- User feedback and ratings storage
- Analytics data persistence
- Theme and configuration storage
- Key Files: actions/conversation_db.py, actions/analytics_store.py
- Database: Chatbot (default, configurable)
- Tables: conversations, messages, feedback, themes, chat_apps, etc.
- Port: 5432

psycopg2-binary (v2.9.9)

- Purpose: PostgreSQL Python adapter/driver
- Usage: Direct database connection and queries
- Key Files: actions/conversation_db.py

■■ SQLAlchemy (<2.0)

- Purpose: Python SQL toolkit and Object Relational Mapper
- Usage:
- Rasa tracker store ORM mapping
- Database connection pooling
- Query building
- Key Files: endpoints.yml (tracker_store configuration)
- Important: Pinned to <2.0 for Rasa 3.6 compatibility

Secrets & Configuration

Infisical SDK (>=1.0.14)

- Purpose: Enterprise secrets management integration
- Usage:
- Retrieve sensitive credentials from Infisical vault
- Manage API keys, tokens securely
- Support multiple environments (dev/prod)
- Key Files: actions/config_manager.py
- Credentials: GLPI tokens, DB passwords, API keys
- Fallback: Local .env file if Infisical unavailable

■ python-dotenv (v1.0.0)

- Purpose: Load environment variables from .env files

- Usage:
- Load database credentials
- Rasa API configuration
- Local development setup
- Key Files: .env (config file), actions/config_manager.py
- Config Keys:

```
DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD
GLPI_API_URL, GLPI_APP_TOKEN, GLPI_USER_TOKEN
RASA_API_URL
```

Machine Learning & Search

Sentence-Transformers (v2.3.1)

- Purpose: Generate semantic embeddings for Knowledge Base search
- Usage:
- Create vector embeddings for KB articles
- Semantic similarity matching
- Hybrid search (semantic + keyword matching)
- Model: all-MiniLM-L6-v2 (384-dim embeddings)
- Key Files: actions/kb_cache_manager.py
- Dependencies: PyTorch or TensorFlow (auto-installed)

System Monitoring & Performance

psutil (v7.2.1)

- Purpose: Cross-platform library for retrieving system & process information
- Usage:
- Monitor CPU usage
- Track memory consumption
- Disk space monitoring
- Process status checks
- Key Files: actions/process_monitor.py
- Admin Panel: System Observability dashboard displays real-time metrics

---

Data Processing

PyYAML (v6.0.1)

- Purpose: Parse and generate YAML files
- Usage:
- Read domain.yml, nlu.yml, stories.yml, rules.yml
- Write configuration files
- Validate YAML syntax
- Key Files: domain.yml, endpoints.yml, multiple action files
- Critical For: Rasa configuration, story/rule definitions

HTTP Client

Requests (v2.31.0)

- Purpose: HTTP library for making API requests
- Usage:
- Call GLPI REST API (tickets, KB articles)
- Communicate with Rasa core/actions
- Call Infisical API
- External webhook calls
- Key Files: actions/glpi_service.py, actions/config_manager.py, custom actions
- Common Endpoints:
- GLPI: http://localhost:8083/apirest.php/*
- Rasa: http://localhost:5005/*
- Infisical: https://app.infisical.com/api/v1/*

Production Server (Optional)

Waitress (v2.1.2)

- Purpose: WSGI application server for production deployment
- Usage: Alternative to Flask dev server for production
- When Used: Production environment (optional, not in dev setup)
- Benefits: Better concurrency handling, stability

Testing (Optional)

■ pytest (v7.4.3)

- Purpose: Python testing framework
- Usage: Unit tests, integration tests

- Test Files: tests/ directory (if present)

Frontend Dependencies

UI Framework

■■ React (v19.2.0)

- Purpose: JavaScript library for building UI components
- Usage:
- Admin dashboard components
- Chat widget UI
- Studio interface
- Admin analytics/config panels
- Key Files: client_ui/src/components/*.jsx, client_ui/src/App.jsx

■ React-DOM (v19.2.0)

- Purpose: React rendering for web browsers
- Usage: Render React components to DOM
- Key Files: client_ui/src/main.jsx

Styling & CSS

■ TailwindCSS (v3.4.17)

- Purpose: Utility-first CSS framework
- Usage:
- Style admin dashboard
- Style chat widget
- Responsive design
- Theme customization
- Key Files: client_ui/tailwind.config.js, client_ui/src/*.css
- Build: Compiled during Vite build

■ PostCSS (v8.4.35)

- Purpose: Tool for transforming CSS
- Usage:
- Process Tailwind directives
- Autoprefixing
- Key Files: client_ui/postcss.config.js

Autoprefixer (v10.4.18)

- Purpose: PostCSS plugin to add vendor prefixes
- Usage:
- CSS compatibility across browsers
- Automatic prefix generation
- Key Files: Part of PostCSS pipeline

■ Tailwind-Merge (v3.4.0)

- Purpose: Merge Tailwind CSS class lists intelligently
- Usage:
- Avoid conflicting Tailwind classes
- Theme override handling
- Key Files: client_ui/src/components/*.jsx (conditional styling)

Icons & UI Components

Lucide-React (v0.561.0)

- Purpose: Beautiful, consistent SVG icon library
- Usage:
- Admin dashboard icons
- Chat widget icons (send, attach, etc.)
- UI iconography
- Icons Used: Send, Paperclip, Minimize, Trash, User, Lock, etc.

Content & Markdown

React-Markdown (v10.1.0)

- Purpose: Render Markdown to React components
- Usage:
- Display markdown formatted bot responses
- Knowledge Base article display
- Rich text formatting
- Key Files: client_ui/src/components/*.jsx

■ Remark-GFM (v4.0.1)

- Purpose: GitHub Flavored Markdown plugin for remark
- Usage:
- Support tables, strikethrough, etc.
- Enhanced markdown features

- Dependencies: React-Markdown plugin

Utilities

■ UUID (v13.0.0)

- Purpose: Generate universally unique identifiers
- Usage:
- Create unique message IDs
- Session identifiers
- Widget instance IDs
- Key Files: client_ui/src/components/ChatWindow.jsx

■■ clsx (v2.1.1)

- Purpose: Utility for constructing className strings
- Usage:
- Conditional CSS class application
- Theme-based styling
- Key Files: client_ui/src/components/*.jsx

Animations

■ Framer-Motion (v12.23.26)

- Purpose: Animation library for React
- Usage:
- Chat widget animations
- Message transitions
- UI element animations
- Smooth page transitions
- Key Files: client_ui/src/components/*.jsx

Build Tools

Core Build System

Vite (v7.2.4)

- Purpose: Next-generation frontend build tool and dev server
- Usage:
- Development server (npm run dev)
- Production bundle creation (npm run build)
- Widget compilation (npm run build:widget)
- Hot module replacement (HMR)
- Key Files:
- client_ui/vite.config.js - Main config
- client_ui/vite.widget.config.js - Widget-specific config
- Output:
- client_ui/dist/ - Admin dashboard
- client_ui/dist-widget/ - Embeddable widget (glpi-chat-widget.js)

@vitejs/plugin-react (v5.1.1)

- Purpose: React plugin for Vite
- Usage:
- JSX transformation
- Fast refresh during development
- Key Files: client_ui/vite.config.js

Code Quality

ESLint (v9.39.1)

- Purpose: JavaScript linter for code quality
- Usage:
- Find and fix problems in code
- Enforce coding standards
- Style consistency
- Key Files: client_ui/.eslintrc.cjs
- Plugins:
- @eslint/js
- eslint-plugin-react-hooks
- eslint-plugin-react-refresh

■ Globals (v16.5.0)

- Purpose: Global variable definitions for ESLint
- Usage:
- Configure ESLint aware of browser/Node globals

Type Safety

@types/react (v19.2.5)

- Purpose: TypeScript type definitions for React
- Usage:
- Type checking in React components
- IDE autocomplete

@types/react-dom (v19.2.3)

- Purpose: TypeScript type definitions for React-DOM
- Usage:
- Type checking for DOM rendering

Server for Widget

http-server (npm, used via script)

- Purpose: Simple HTTP server for static files
- Usage:
- Serve the compiled widget (dist-widget/)
- Allow embed testing on different ports
- Command: npx -y http-server dist-widget -p 8181 --cors -c-1
- Port: 8181 (widget serving)
- Flags: CORS enabled, no cache

External Systems & Services

GLPI (ITSM)

■ GLPI (External System)

- Purpose: IT Service Management platform
- Integration Method: REST API
- API Endpoint: http://localhost:8083/apirest.php
- Operations:
- Create/read/update tickets
- Knowledge base articles
- User authentication
- Entity/department management

- Authentication: App Token + User Token
- Key Files: actions/glpi_service.py
- Tokens Stored In: Infisical or .env

Infisical Vault

Infisical (External Secrets Service)

- Purpose: Enterprise secrets vault
- Integration Method: HTTP API + SDK
- API Endpoint: https://app.infisical.com/api/v1/
- Operations:
- Retrieve API tokens
- Store database credentials
- Environment-based secrets
- Fallback: Local .env if Infisical unavailable
- Key Files: actions/config_manager.py
- Credentials: Universal Auth client ID & secret

Rasa Services

■ Rasa Core (Internal Service)

- Purpose: NLU and dialogue management engine
- Port: 5005
- API Endpoints:
- /webhooks/rest/webhook - Chat messages
- /model/parse - NLU analysis
- /domain - Domain retrieval
- /conversations/{id}/tracker - Conversation history

■ Rasa Action Server (Internal Service)

- Purpose: Execute custom actions
- Port: 5055
- API Endpoints:
- /webhook - Receive action requests
- /health - Health check

Infrastructure & DevOps

Containerization

Docker (Container Runtime)

- Purpose: Package application in containers
- Key Files: Dockerfile (if present), docker-compose.yml, docker-compose.infisical.yml
- Containers:
- Rasa Core container
- Action Server container
- Admin API container
- PostgreSQL container
- GLPI container
- Nginx container

Docker Compose (Container Orchestration)

- Purpose: Define and run multi-container applications
- Configs:
- docker-compose.yml - Base configuration
- docker-compose.infisical.yml - With Infisical integration
- Services: Rasa, Actions, Admin, Database, Nginx, GLPI

Reverse Proxy & Load Balancing

Nginx (Reverse Proxy)

- Purpose: HTTP reverse proxy and load balancer
- Usage:
- Unified entry point (port 80/443)
- Route requests to backend services
- SSL/TLS termination
- Static file serving
- Routes:
- / → Admin API (port 80)
- /webhooks/* → Rasa Core (port 5005)
- /actions/* → Action Server (port 5055)
- Key Files: Nginx config (in deployment)

Version Control

Git (Version Control System)

- Purpose: Source code version control
- Usage:
- Track code changes
- Branching and merging
- Collaborative development
- Key Files: .git/, .gitignore

Runtime

■ Python 3.x (Runtime)

- Purpose: Python interpreter for backend
- Version: 3.8+ (recommended 3.10+)
- Package Manager: pip
- Virtual Environment: venv/conda

Node.js (JavaScript Runtime)

- Purpose: JavaScript runtime for frontend build & serving
- Package Manager: npm
- Key Files: package.json, package-lock.json

Development Tools & Scripts

Shell Scripts (Windows)

Batch/PowerShell Scripts

- start_services.bat - Start all services
- stop_services.bat - Stop all services
- restart_services.bat - Restart all services
- restart_admin_api.bat - Restart admin API
- setup.bat - Initial setup
- backup_database.bat - Backup PostgreSQL
- restore_database.bat - Restore from backup
- health_check.bat - Health check script

PowerShell Scripts

- start_all.ps1 - Start services (PS version)

• test_departments.ps1 - Department testing

Summary Table

Library	Version	Type	Purpose	Key File(s)
---	---	---	---	---
rasa	3.6.20	Framework	NLU + Dialogue	domain.yml, nlu.yml, stories.yml
rasa-sdk	3.6.2	SDK	Custom Actions	actions/actions.py
flask	3.0.0	Framework	Admin API	actions/admin_api.py
flask-cors	4.0.0	Middleware	CORS Support	actions/admin_api.py
psycopg2-binary	2.9.9	Driver	PostgreSQL	actions/conversation_db.py
sqlalchemy	<2.0	ORM	Database Access	endpoints.yml
pydantic	1.10.9	Validation	Data Validation	config models
infisicalsdk	>=1.0.14	SDK	Secrets	actions/config_manager.py
python-dotenv	1.0.0	Config	.env Loading	.env
sentence-transformers	2.3.1	ML	KB Embeddings	actions/kb_cache_manager.py
psutil	7.2.1	Monitoring	System Stats	actions/process_monitor.py
pyyaml	6.0.1	Parser	YAML Processing	All config files
requests	2.31.0	HTTP	API Calls	actions/glpi_service.py
waitress	2.1.2	Server	WSGI Server	(optional)
pytest	7.4.3	Testing	Unit Tests	tests/
react	19.2.0	Framework	UI Components	client_ui/src/
react-dom	19.2.0	Library	DOM Rendering	client_ui/src/
vite	7.2.4	Build Tool	Build System	client_ui/vite.config.js
tailwindcss	3.4.17	CSS	Styling	client_ui/tailwind.config.js
framer-motion	12.23.26	Animation	UI Animation	client_ui/src/components/
lucide-react	0.561.0	Icons	Icon Library	client_ui/src/components/
uuid	13.0.0	Utility	ID Generation	client_ui/src/components/

react-markdown	10.1.0	Component	Markdown Rendering	client_ui/src/components/

Dependency Graph

```
Project-IQ
■
■■■ Backend Services
■   ■■■ Rasa Core (v3.6.20)
■   ■   ■■■ TensorFlow/PyTorch
■   ■   ■■■ Sanic (async web)
■   ■   ■■■ YAML (v6.0.1)
■   ■
■   ■■■ Action Server (rasa-sdk v3.6.2)
■   ■   ■■■ Flask (v3.0.0)
■   ■   ■■■ Requests (v2.31.0)
■   ■   ■■■ psycopg2 (v2.9.9)
■   ■   ■■■ SQLAlchemy (<2.0)
■   ■   ■■■ sentence-transformers (v2.3.1)
■   ■   ■■■ psutil (v7.2.1)
■   ■   ■■■ Infisical SDK (>=1.0.14)
■   ■   ■■■ python-dotenv (v1.0.0)
■   ■
■   ■■■ Admin API (Flask v3.0.0)
■       ■■■ Flask-CORS (v4.0.0)
■       ■■■ Werkzeug (v3.0.1)
■       ■■■ Static Files (HTML/CSS/JS)
■
■■■ Frontend Stack
■   ■■■ Build System: Vite (v7.2.4)
■   ■   ■■■ @vitejs/plugin-react (v5.1.1)
■   ■   ■■■ TailwindCSS (v3.4.17)
■   ■       ■■■ PostCSS (v8.4.35)
■   ■       ■■■ Autoprefixer (v10.4.18)
■   ■
■   ■■■ UI Framework: React (v19.2.0)
■   ■   ■■■ React-DOM (v19.2.0)
■   ■   ■■■ Framer-Motion (v12.23.26)
■   ■   ■■■ Lucide-React (v0.561.0)
■   ■   ■■■ UUID (v13.0.0)
■   ■   ■■■ clsx (v2.1.1)
■   ■   ■■■ Tailwind-Merge (v3.4.0)
■   ■   ■■■ React-Markdown (v10.1.0)
■   ■   ■■■ Remark-GFM (v4.0.1)
■   ■
■   ■■■ Code Quality: ESLint (v9.39.1)
■       ■■■ @eslint/js
■       ■■■ eslint-plugin-react-hooks
■       ■■■ eslint-plugin-react-refresh
■
■■■ External Services
■   ■■■ GLPI (REST API)
■   ■■■ Infisical (Secrets Vault)
■   ■■■ PostgreSQL (Database)
■   ■■■ Nginx (Reverse Proxy)
■
■■■ DevOps
    ■■■ Docker (Containerization)
    ■■■ Docker Compose (Orchestration)
    ■■■ Git (Version Control)
```

Licensing Summary

Dependency	License	Type
----------	---------	------
Rasa	Apache 2.0	Open Source
Flask	BSD-3-Clause	Open Source
React	MIT	Open Source
TailwindCSS	MIT	Open Source
PostgreSQL	PostgreSQL	Open Source
Docker	Apache 2.0	Open Source
Nginx	BSD-2-Clause	Open Source
sentence-transformers	Apache 2.0	Open Source
Framer-Motion	MIT	Open Source

Installation Commands

Backend Dependencies

```
pip install -r requirements.txt
```

Frontend Dependencies

```
cd client_ui
npm install
```

Build Frontend

```
npm run build
npm run build:widget
```

Environment Requirements

Component	Requirement

| ---------- | ---------- |
OS	Windows, Linux, macOS
Python	3.8+ (recommend 3.10+)
Node.js	16+ (recommend 18+)
RAM	4GB minimum (8GB+ recommended)
Disk	5GB minimum
Database	PostgreSQL 12+
Docker	v20+ (for containerized deployment)

Critical Version Constraints

■■ Important Pinning Notes

1. SQLAlchemy <2.0
   - Rasa 3.6 is not compatible with SQLAlchemy 2.0+
   - Must be pinned to v1.x series
2. Rasa 3.6.20
   - Requires specific versions of dependencies
   - TensorFlow 2.12+ or PyTorch 1.12+
   - Compatible with Python 3.8-3.11
3. React 19.2.0
   - Modern React with concurrent features
   - Requires Node 16+
4. TailwindCSS 3.4.17
   - Requires PostCSS 8+
   - Node 14+

Monitoring & Health Checks

Health Check Endpoints

Service	Endpoint	Port

| --- | --- | --- |
Rasa Core	http://localhost:5005/health	5005
Action Server	http://localhost:5055/health	5055
Admin API	http://localhost:80/health or http://localhost:8181/health	80/8181
PostgreSQL	Connection via psycopg2	5432

Troubleshooting

Common Dependency Issues

Issue	Cause	Solution
Rasa import error	TensorFlow not installed	pip install tensorflow
Flask port conflict	Port 80 already in use	Change port or stop conflicting service
React build fails	Node version too old	Upgrade to Node 16+
DB connection error	PostgreSQL not running	Start PostgreSQL service
CORS errors	Flask-CORS not configured	Check admin_api.py CORS setup
Sentence-transformers slow	Model not cached	First run downloads 500MB+ model

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Additional Resources

- [Rasa Documentation](https://rasa.com/docs)
- [Flask Documentation](https://flask.palletsprojects.com)
- [React Documentation](https://react.dev)
- [TailwindCSS Documentation](https://tailwindcss.com)
- [PostgreSQL Documentation](https://www.postgresql.org/docs)
- [Docker Documentation](https://docs.docker.com)

Document Status: ■ Complete